

When Finite Differences Flip the Sign of Second-Order Generative Geometry: A Non-Recoverability Result and an Exact Instrument

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Abstract

Studies of the differential geometry of generative models increasingly measure *second-order* quantities—curvature of the generated manifold and the terms that enter geodesics. We report a concrete, reproducible hazard in how these are computed. On Stable Diffusion, a representative second-order geometric measurement—the rank correlation between the first-order singular value σ of the latent-to-image Jacobian and a second-order curvature ratio ρ along the leading directions—comes out *with opposite sign* depending only on the numerical method: the exact forward-mode estimate gives a consistent strong negative correlation (Spearman -0.73 , 5/6 seeds ≤ -0.82), whereas a finite-difference (FD) estimate of the same quantity is sign-unstable and averages -0.18 . The first-order Jacobian, by contrast, is recovered by finite differences to 2.4%. We explain this asymmetry: central FD for a *second* derivative is *non-recoverable*—truncation and floating-point cancellation trade off so that no step size beats a $\sqrt{\varepsilon_{\text{mach}}}$ relative-error floor, which in the single precision generators use is large—while FD for a first derivative is well conditioned. On a StyleGAN generator the naive FD curvature estimate misses the exact value by $\geq 45\%$ at every fixed step, in *both* single and double precision (validated against the exact instrument by the first-order FD, which agrees to 2.4%); precision does not rescue it. We provide the exact constant-memory instrument—composed forward-mode automatic differentiation, generator-agnostic and extending to any order—and survey how the literature computes second-order geometry, finding that the catastrophic explicit- $/\varepsilon^2$ regime is the natural first thing to reach for but is, fortunately, avoided by most published methods. Our contribution is the precise account of when finite differences fail for second-order generative geometry, the practitioner-facing first/second-order asymmetry, and a drop-in exact replacement.

1 Introduction

Let $G : \mathbb{R}^d \rightarrow \mathbb{R}^{3HW}$ be a trained generator. Treating $\{G(z)\}$ as a manifold pulled back through G , a productive line of work studies its geometry: the pullback metric $M_z = J_z^\top J_z$ and its singular vectors (the local latent basis), geodesics between latent codes, and the curvature of the generated manifold (Arvanitidis et al., 2018; Shao et al., 2018; Chen et al., 2018; Park et al., 2023; Saito & Matsubara, 2025). First-order objects (J_z , M_z) are routine to compute. The *second-order* objects—curvature, the second-order pushforward $\partial^2 G / \partial \alpha^2$ along a path, the metric derivative $\partial_z M_z$ that appears in the geodesic ODE—where numerical practice matters and, we show, can silently flip a result.

Our entry point is an observation, not a theorem (Section 2). A standard second-order measurement on Stable Diffusion returns the *opposite sign* depending only on whether the second derivative is taken by finite differences or by exact forward-mode AD. The exact method finds a clean, seed-consistent relationship; the finite-difference method, applied to the very same quantity, is sign-unstable and washes it out. A practitioner choosing the natural finite-difference route would draw the wrong qualitative conclusion.

The explanation is an asymmetry that is elementary once stated but, in our experience, not widely appreciated in this applied setting (Section 3): a central finite difference is well conditioned for a *first* derivative but

non-recoverable for a *second* derivative, because for the second derivative truncation error and floating-point cancellation cannot be made small at the same step size, leaving a relative-error floor of order $\sqrt{\varepsilon_{\text{mach}}}$. In the single precision that GANs and diffusion U-Nets run in, that floor is large; and on a strongly nonlinear, non-smooth generator it is large at *every* step and in *both* precisions (Section 4: $\geq 45\%$ on StyleGAN, fp32 and fp64 alike).

We make three contributions: (i) the reproducible sign-inversion phenomenon and the first/second-order asymmetry that governs it, as concrete guidance; (ii) an exact, constant-memory instrument for the higher-order pushforward via composed forward-mode AD, which removes the hazard at the cost of one extra forward pass per order; and (iii) a survey of how the field computes second-order geometry, locating where the trap is and is not. We are explicit about scope (Section 7): we do not claim any published result is wrong—indeed we find most published methods route around the catastrophic regime—we characterize a real numerical hazard and supply the fix.

2 The sign-inversion phenomenon

On a Stable-Diffusion h-space pipeline we take a base latent, form the first-order Jacobian J of the (truncated) denoising map along K random h-space probes, and along its top- k right-singular directions u_i measure a second-order curvature ratio

$$\rho_i = \frac{|\partial_\alpha^2 G(z + \alpha u_i)|}{|\partial_\alpha G(z + \alpha u_i)|} \Big|_{\alpha=0},$$

the magnitude of the second-order pushforward relative to the first. We then ask a standard question—does curvature co-vary with the first-order singular value σ_i ?—via $\text{Spearman}(\sigma, \rho)$, computing the second derivative in ρ two ways: exact composed JVP (Section 5) and a finite difference of the JVP (the natural way a practitioner would estimate it).

Over 6 seeds (top-12 directions each), the two methods disagree in *sign* (Table 1). The exact instrument yields a strong negative correlation on five of six seeds (mean -0.73 ; the five clustered in $[-0.94, -0.82]$, with seed 4 a near-zero outlier at -0.03). The finite-difference estimate of the *same* ratio is sign-unstable across seeds and averages -0.18 : it does not merely add noise, it destroys the relationship. The exact estimate is more negative than the FD estimate on *all six* seeds (Wilcoxon signed-rank $p = 0.031$, the floor attainable at $n=6$), and the FD estimate is sign-unstable at *every* step in the grid $\{0.1, 0.05, 0.01, 0.001\}$ (means -0.14 to -0.34): no step reproduces the clean -0.73 , so the disagreement is not an artifact of the chosen step. A study relying on the finite-difference route would report “no clear relationship,” the opposite of what the exact second-order estimate finds.

seed	0	1	2	3	4	5	mean
exact composed JVP	−0.85	−0.83	−0.92	−0.94	−0.03	−0.84	−0.73
FD-of-JVP($\varepsilon=0.05$)	+0.18	−0.55	−0.78	−0.40	+0.55	−0.07	−0.18

Table 1: $\text{Spearman}(\sigma, \rho)$ on Stable Diffusion, per seed. The exact second-order estimate is consistently strongly negative; the finite-difference estimate of the identical quantity changes sign across seeds and averages near zero. (ρ is a measurement we define to expose the method dependence; the point is that the two numerical routes disagree on its sign, not the specific value of any prior work.)

3 Why: finite differences are non-recoverable for a second derivative

For smooth scalar f the central second difference is $D_\varepsilon^2 f = (f(\alpha + \varepsilon) - 2f(\alpha) + f(\alpha - \varepsilon))/\varepsilon^2$, with

$$D_\varepsilon^2 f - f''(\alpha) = \underbrace{\frac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4)}_{\text{truncation}} + \underbrace{O(\varepsilon_{\text{mach}} |f|/\varepsilon^2)}_{\text{cancellation}}.$$

The two terms cannot be made small together: minimizing their sum gives an optimal step $\varepsilon^* \sim \varepsilon_{\text{mach}}^{1/4}$ and a relative-error floor (dividing the absolute error above by $|f''|$, assuming $f^{(4)}$ bounded, $f'' \neq 0$, and $|f|$ normalized to $O(1)$) of $2\sqrt{|f^{(4)}f|/|f''|} \cdot \varepsilon_{\text{mach}}^{1/2}$. The *exponent* $\frac{1}{2}$ is intrinsic to the operation; the constant carries the generator’s nonlinearity $|f^{(4)}f|/f''^2$, which is large for a deep network. In double precision the floor is $\sim 10^{-8}$ in the benign (constant ≈ 1) case; in the single precision generators use ($\varepsilon_{\text{mach}} \approx 1.2 \times 10^{-7}$ for fp32) it is $\sim 3 \times 10^{-4}$ benign and far worse for a strongly nonlinear G (Section 4: $\geq 45\%$). No choice of ε removes the floor. Because the floor is a property of the finite-difference *operation* in fixed precision, not of any architecture, the hazard is general; we demonstrate its magnitude on two generator families (StyleGAN and a latent-diffusion U-Net).

The asymmetry. The *first* central difference $(f(\alpha+\varepsilon)-f(\alpha-\varepsilon))/2\varepsilon$ has truncation $O(\varepsilon^2)$ and cancellation $O(\varepsilon_{\text{mach}}|f|/\varepsilon)$, giving a much milder floor $\sim \varepsilon_{\text{mach}}^{2/3}$ and a wide usable window. This is why first-order Jacobians are safely finite-differenced while second-order curvature is not—and it is the control we use to validate our instrument (Sections 2, 4): wherever the first-order FD agrees with exact JVP, the pipeline is correct and any second-order disagreement is genuine.

Analytic referee. Let $f(\alpha) = [e^{0.7\alpha}, \sin 3\alpha, (\alpha + 0.4)^3, \tanh 2\alpha, \alpha^2 \cos \alpha]$, with closed-form derivatives. At $\alpha = 0.3$ in double precision (Table 2) the central-FD second-order relative error is U-shaped, bottoming at 8.6×10^{-9} near $\varepsilon = 10^{-4}$ and diverging to 0.29 by $\varepsilon = 10^{-8}$; the composed JVP attains 2.5×10^{-17} —FD’s best is 3.5×10^8 times worse.

ε	central FD 2nd-order rel. err.	
10^{-1}	7.9×10^{-3}	(truncation)
10^{-2}	8.0×10^{-5}	
10^{-4}	8.6×10^{-9}	(best)
10^{-6}	3.0×10^{-5}	
10^{-8}	2.9×10^{-1}	(cancellation)
composed JVP	2.5×10^{-17}	(exact)

Table 2: Second-order accuracy on the analytic toy at $\alpha = 0.3$, double precision. Finite differences never approach the exact composed JVP.

4 Quantification on a real generator

The toy fixes ideas; on a real generator the failure is both worse and broader than cancellation alone. On a StyleGAN-bedroom generator (256×256) we form the naive FD curvature estimate a practitioner would write, $D_\delta^2 G = (G(z+\delta u) - 2G(z) + G(z-\delta u))/\delta^2$, and the exact composed JVP $\partial_\alpha^2 G(z+\alpha u)$, over 6 directions (3 attributes, 3 random) × 16 samples, sweeping δ (Table 3). The exact curvature is a definite, step-free number (mean 1.17). Holding a *single* step across all samples—what a practitioner does—the FD mean relative error never drops below 45.2% (best step $\delta = 3$, bracketed on both sides); it is 267% at $\delta \sim 0.3$ and explodes as $\delta \rightarrow 0$. Even allowing an *oracle* per-sample step the median best error is still $\sim 9\%$ (vs. the composed JVP’s 0). **Precision does not rescue it:** rerunning the entire sweep in fp64 leaves the floor unchanged (47.1% in both fp32 and fp64; Table 3 note). The mechanism is not cancellation alone: at usable steps the error is truncation (precision-blind); at tiny steps fp32 adds catastrophic cancellation, but fp64—which removes cancellation—instead exposes the generator’s *non-smoothness* (LeakyReLU kinks), where the chord-scale second difference does not converge to the pointwise pushforward (fp64 error grows to $> 3000\%$ as $\delta \rightarrow 0$). So on a real generator *no step and no precision* recovers the exact second-order pushforward, which the composed JVP returns directly. The control: the *first-order* central difference of the same pipeline agrees with the exact first-order JVP to 2.4%, so the instrument and pipeline are correct and the second-order failure is genuine, not a bug.

step δ	$ D_\delta^2 G $	2nd-order rel. err.	1st-order rel. err.
8	7.1×10^{-3}	74.5%	63.8%
3	2.7×10^{-2}	45.2% (floor)	44.1%
1	9.9×10^{-2}	106.4%	26.2%
0.3	3.7×10^{-1}	266.9%	12.5%
0.1	1.06	622.8%	5.5%
0.02	4.05	$6.8 \times 10^3\%$	2.4% (1st-order min)
10^{-3}	4.09×10^2	$2.0 \times 10^6\%$	139%
composed JVP	1.17	exact (step-free)	—

Table 3: StyleGAN-bedroom, fp32. Columns are means over 6 directions $\times 16$ samples at each fixed step; rel. err. is the mean per-direction relative error, while $|D_\delta^2 G|$ is the aggregate magnitude (not its direct source). The naive FD second difference never gets within 45% of the exact curvature at any fixed step; the first-order FD of the same code agrees to 2.4%, validating the instrument. Rerunning in fp64 leaves the best-step floor unchanged (47.1% fp32 vs. 47.1% fp64), so the floor is not a single-precision artifact—it is truncation at usable steps and generator non-smoothness at tiny steps.

5 The exact instrument: composed forward-mode AD

Forward-mode AD evaluates, for a map f and tangent v , the Jacobian–vector product $\partial f(x)[v]$ alongside $f(x)$ in one pass without forming the Jacobian. With $f = G$, $x = \gamma(\alpha)$, $v = u$ this gives the exact first-order pushforward. The second-order term is the directional derivative of the first, so composing forward mode with itself, $\partial_\alpha^2 G(\gamma(\alpha)) = \partial_\alpha [\partial G(\gamma(\alpha))[u]][u]$, is exact and is evaluated by nesting two JVP calls (`torch.func.jvp` of a function that returns a JVP). No Jacobian is stored, so memory stays at one forward pass; the same nesting yields any order in a constant number of passes. The instrument is generator-agnostic (we ran it unmodified on StyleGAN1/2 and a latent-diffusion U-Net) and, unlike reverse-mode Hessian materialization (memory $O(3HW)$), its cost is independent of output dimension.

6 Survey: where the trap is, and is not

Table 4 classifies representative latent-geometry papers by how they obtain second-order/geodesic quantities. The explicit $/\varepsilon^2$ second-order finite difference—the catastrophic regime of Section 3—is the natural first thing to reach for, but most published methods avoid it: they keep the metric first-order, replace the geodesic ODE with a discretized energy whose *exact* gradient is a well-conditioned discrete Laplacian (not a $/\varepsilon^2$ curvature), use analytic Christoffel symbols, or differentiate the energy with autodiff (Yang et al., 2018); first-order constructions (Park et al., 2023; Chen et al., 2018; Kwon et al., 2023) never touch a second derivative. The genuinely exposed practice is first-order FD of a network derivative (e.g. a score-Jacobian–vector product), which is only mildly inaccurate, and—most relevant here—the direct $/\varepsilon^2$ curvature estimate that a practitioner adds on top of an existing pipeline, as in our Section 2 measurement. The contribution of the survey is to draw this line precisely. The reason the field is mostly safe is structural—the canonical formulations are first-order metrics or energy minimizations *by construction*, which never form an explicit second-order finite difference. But that safety is a property of the current formulations, not a law: as curvature- and geodesic-level analysis of diffusion models grows (e.g. Saito & Matsubara (2025), and any second-order extension of the first-order bases of Park et al. (2023)), the $/\varepsilon^2$ estimate becomes the obvious next step. We intend the line in Table 4, the asymmetry, and the drop-in instrument as forward-looking guardrails for that emerging work, not as a retrospective indictment.

Guidance. The practical rule is short. *First-order* quantities (Jacobians, pullback metrics, first-order pushforwards) may be finite-differenced; choose a step near $\varepsilon_{\text{mach}}^{1/3}$ and expect a few digits. *Second or higher-order* quantities (curvature, the second-order pushforward, the metric derivative / Christoffel symbols, anything with a $/\varepsilon^2$) should *not* be finite-differenced: in single precision no step gives even one reliable digit on a deep generator (Section 4), and the error can flip the sign of a downstream conclusion (Section 2). Use composed

Paper	2nd-order quantity	how obtained
Shao et al. (2018)	discretized geodesic energy	exact discrete-energy gradient (Laplacian), coarse T ; no / ε^2
Saito & Matsubara (2025)	geodesic (energy)	first-order score-differences $\ s(x_{i+1}) - s(x_i)\ ^2$ (no $/\varepsilon^2$); JVP backend unspecified
Arvanitidis et al. (2018)	geodesic ODE (Christoffel)	analytic ∂M , numeric ODE integ.
Yang et al. (2018)	geodesic energy	autodiff
Park et al. (2023)	none (first-order basis)	autodiff power iteration
Chen et al. (2018)	none (first-order metric)	—
Kwon et al. (2023)	none (first-order)	CLIP-grad + MLP

Table 4: How representative latent-geometry papers obtain second-order quantities. The explicit $/\varepsilon^2$ second-order finite difference (Sections 3, 4) is largely avoided in published work; it is nonetheless the natural choice for a practitioner adding a curvature measurement, which is where the hazard bites.

forward-mode AD instead—it is exact, costs one extra forward pass per order, and adds no memory. A free diagnostic comes with it: compare the first-order finite difference to the first-order JVP; if they disagree, the pipeline is wrong, and if they agree, any second-order disagreement is real, not numerical.

7 Scope, limitations, and what we do not claim

This is a numerical-methods contribution with a survey and demonstrations. **(1)** We do *not* claim any surveyed paper reached a false conclusion; on the contrary, the survey shows most route around the catastrophic regime. The sign inversion of Section 2 is on a quantity *we* define to expose method dependence, not a number from prior work. **(2)** The non-recoverability result is specifically about *second*-order finite differences; first-order finite differences (e.g. score-JVPs) are well conditioned, and we say so. **(3)** The instrument is a thin, exact wrapper around forward-mode AD; its value is in routine exact higher-order pushforward and in the precise account of when the finite-difference alternative fails, not in algorithmic novelty. **(4)** We do not propose a downstream task win: this is a measurement-correctness result, not a demonstration that exact curvature improves an application. Whether accurate second-order geometry changes downstream conclusions is left to future work.

8 Reproducibility

We release the exact-instrument module, the analytic referee, and the StyleGAN and Stable-Diffusion scripts with fixed seeds and the emitted `metrics.json` for every table. All experiments run on a single 8 GB GPU.

References

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